AEGIS V2 — Canonical Master Plan v2.2

Date: 2026-03-24 05:00 UTC (all post-fix truth, roadmap integrated) **Status**: COMPLETE — 35 sections + 2 appendices. All sections reflect POST-FIX state only. **Supersedes**: AEGIS_V2_CANONICAL_MASTER_PLAN_20260324.md and ALL prior plan docs **Source of truth hierarchy**: Code > Runtime > Config > This plan > Old docs **Verification method**: 4 parallel agents read all code + config + EC2 runtime + logs

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1. Executive Summary

AEGIS V2 is a **paper-trading prototype** running on EC2 c7i-flex.large (4GB RAM, 19GB disk at 80% usage). It processes live IBKR market data through a Rust execution engine (32,603 LOC, 51 modules) with a Python signal brain (1,863 LOC bridge + 6,000+ LOC supporting files). `IS_LIVE=false` is hardcoded — the system cannot place real orders.

Current performance: 48 nightly-tracked trades (since 2026-03-18), 35.4% WR, -£6.79 cumulative P&L, PF=0.0 (bug — fixed this session, not yet run through nightly). Only VanguardSniper has produced trades (33 of the 48). All other strategies have 0 production signals.

CRITICAL RUNTIME ISSUE DISCOVERED THIS SESSION: The system is currently **generating signals but ALL are being vetoed** by RiskArbiter CHECK 10 (ConfidenceBelowFloor). Signals arrive at confidence=49.0 but the floor (from dynamic_weights.toml) is 65. The bridge's base confidence in non-LSE markets is too low to pass. Only 11 of 50 configured tickers are actively receiving data — the rest hit IBKR's tick-by-tick limit. The system is effectively **dead** outside LSE market hours.

This session (2026-03-24 session 2): Sprint S1 (paper fill audit) and S2 (PF bug fix) completed. Plan expanded from 22 to 35 sections with full evidence-based detail.

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2. Current Verified System State

Fact	Value	Evidence	Verified
Branch	`feat/tier-system-enhancements-full` @ `d17d9bd`	`git log --oneline -1`	Yes
Containers	3 healthy (aegis-v2, ib-gateway, redis)	`docker ps` on EC2	Yes
Disk	80% (3.9GB free / 19GB)	`df -h /` on EC2	Yes
IB Gateway	Connected, 2FA approved	Subscription active	Yes
Market data subscriptions	200 mkt + 102 L1 attempted, but tick-by-tick limit hit for many	Engine logs	Yes
Active tickers receiving data	**11** (not 50)	`active_watchlist.json` on EC2	Yes
Python Bridge	Running, producing signals	SIGNAL_ARRIVED logs	Yes
Signal outcome	**ALL vetoed** (ConfidenceBelowFloor, conf=49 < floor=55)	VETO logs	Yes
Ouroboros	FROZEN (observe_only=true)	config.toml	Yes
Gemini API key	**COMMENTED OUT in .env** — Gemini crons fail silently	`.env` inspection	Yes
Claude	CLI at /usr/bin/claude, shadow mode (non-blocking)	entrypoint.sh	Yes

Strategies deployed	6 sources + TypeA-F classification in bridge.py	Code audit	Yes
Strategies producing signals	0 right now (all vetoed)	Runtime logs	Yes
Strategies with historical trades	1 (VanguardSniper: 33), 1 (TypeE: 3 classified)	WAL data	Yes
Equity	£10,000	system_memory.json	Yes
Open positions	0	Heartbeat logs	Yes
dynamic_weights.toml	**STALE** — WR=79.2% from 20 trades, confidence_floor=65	File content	Yes
Nightly pipeline	Runs 04:50 UTC via flock'd script	Crontab	Yes
system_memory PF	0.0 (BUG — fix deployed, awaits next nightly)	system_memory.json	Yes
Risk CHECKs	**27 active** (not 33 as previously documented)	risk_arbiter.rs audit	Yes
Time-stop	**ABSENT** despite config.toml having `[exit_time_stop] enabled=true`	exit_engine.rs audit	Yes
Cron jobs	**35 scheduled jobs** in crontab	Crontab audit	Yes
Config sections	**30 sections** in config.toml	Full file read	Yes

3. Authoritative Source-of-Truth Hierarchy

Rank	Source	What it governs
1	Rust code (`rust_core/src/`)	Execution, risk, exits, WAL, broker
2	Python code (`python_brain/`)	Signals, sizing, classification, learning
3	Config files (`config/`)	Parameters, thresholds, universe
4	EC2 runtime state	What's actually deployed and running
5	WAL events (`/app/events/`)	Trade history, P&L
6	system_memory.json	Nightly-aggregated performance
7	This plan (v2.0)	Architecture decisions, roadmap
8	Old plan docs (docs/archive/)	Historical context only, NOT authoritative

4. Actual System Architecture

```
IBKR Gateway (aegis-ib-gateway:4003)
  │
  ├─ Market data ticks (200 mkt subs + 102 L1 attempted)
  │    └─ PROBLEM: tick-by-tick limit hit → only ~11 tickers get real data
  │
  └─ Rust engine.rs (process_tick_with_signal)
       │
       ├─ Pre-processing: FX, GBX→GBP, gap detection, GARCH, Kalman, regime
       │
       ├─ EXIT path (runs first, always — even in Dark mode)
       │    └─ exit_engine.rs: Chandelier 5-rung, EOD flatten, volume exhaustion
       │         └─ NO TIME-STOP (config says enabled but code doesn't implement)
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        │
        ├─ ENTRY path (gated by session, mode, exchange hours, 15+ filters)
        │   ├─ Python bridge.py (stdin/stdout JSON IPC, 5-stage pipeline)
        │   │   ├─ Stage 0: Blackout/blacklist/warmup gates
        │   │   ├─ Stage 1: Indicator computation (RVOL, Hurst, ADX, VWAP, IBS)
        │   │   ├─ Stage 2: Quality gates (spread, VWAP extension, STS)
        │   │   ├─ Stage 3: 6 signal sources (VanguardSniper, Orchestrator, IBS, VolExp, ORB, GapFade)
        │   │   ├─ Stage 4: Adjustments + TypeA-F classification + Claude shadow
        │   │   └─ Stage 5: Best signal output
        │   │
        │   └─ RiskArbiter.evaluate() → 27 deterministic CHECKs
        │       └─ PROBLEM: CHECK 10 vetoing ALL signals (conf=49 < floor=55)
        │
        └─ WAL events → /app/events/current.ndjson
            └─ Nightly pipeline (04:50 UTC, flock'd)
                ├─ nightly_v6.py → daily metrics + recommendations
                ├─ config_writer.py → dynamic_weights.toml (FROZEN — observe_only)
                ├─ persistent_memory.py → system_memory.json (PF FIX DEPLOYED)
                ├─ Claude forensic review (04:53 UTC)
                └─ Fill quality + post-trade diagnostics (04:54-55 UTC)

Supporting infrastructure:
    ├─ Redis (aegis-redis:6379) — state journal, password-protected
    ├─ Telegram — kill switch, alerts, heartbeat, reports
    ├─ Google Sheets — trade sync every 5 min
    ├─ ticker_selector.py — universe rotation every 2h (15min during market hours)
    ├─ Gemini scanner — universe curation every 2h (BROKEN — API key not in .env)
    └─ 35 total cron jobs via supercronic
```

5. Runtime Ownership Map

Concern	Owner	Authority	File(s)	Notes
Tick processing	Rust engine.rs	FINAL	engine.rs:891+	100ms main loop
Entry risk gating	Rust risk_arbiter.rs	FINAL, deterministic	risk_arbiter.rs	27 CHECKs
Exit decisions	Rust exit_engine.rs	FINAL, Chandelier	exit_engine.rs	No time-stop
Signal generation	Python bridge.py	6 sources, best-by-confidence	bridge.py	Hot path, every tick
Entry type classification	Python bridge.py Stage 4	classify_entry_type()	bridge.py:601+	Python only (Rust version unused)
Position sizing	Python kelly_12factor.py	12-factor multiplicative	kelly_12factor.py	Called per signal
Nightly learning	Python nightly_v6.py	Analysis only (FROZEN)	nightly_v6.py	04:50 UTC cron
Config generation	Python config_writer.py	FROZEN (observe_only)	config_writer.py	04:51 UTC cron
Universe curation	Python ticker_selector.py	Writes active_watchlist.json	ticker_selector.py	Every 2h
Signal challenge	Claude curator	Shadow only, non-blocking, fail-open	claude_curator.py	>55 confidence only
Universe AI curation	Gemini	BROKEN (API key commented out)	gemini_scanner.py	Every 2h (silently failing)

6. Rust / Python / Config / Runtime Authority Boundaries

Decision	Rust	Python	Config	Runtime
"Should we enter?"	CHECK 1-27 veto	Signal generation + confidence	Thresholds in config.toml	dynamic_weights overrides
"What price?"	tick.ask (entry), tick.bid (exit)	N/A	marketable_limit_buffer_pct	N/A
"How many shares?"	qty = notional / ask	Kelly 12-factor sizing	kelly caps, sizing params	Half-Kelly for <250 trades
"When to exit?"	Chandelier rung + EOD	N/A	rung_pct, ATR mults	adaptive chandelier
"Which tickers?"	Universe from watchlist	ticker_selector scoring	contracts.toml	active_watchlist.json
"What confidence?"	Floor from config	Base from strategy + adjustments	confidence_floor	adaptive from dynamic_weights

Conflict: dynamic_weights.toml sets confidence_floor=65 but Python bridge generates signals at conf=49 for non-LSE tickers. This effectively blocks ALL non-LSE trading.

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7. Strategy Inventory and Classification

Strategy	Canonical Name	Status	Python path	Config	Observed
VanguardSniper	Momentum	LIVE	bridge.py:1302	brain/config.py	YES
Orchestrator S17	VWAP Dip Buy	LIVE	bridge.py:1318→autonomous_orchestrator	strategies.toml	NO
Orchestrator S18	Gap Fade	LIVE	bridge.py:1318→autonomous_orchestrator	strategies.toml	NO
Orchestrator S19	RSI(2)/IBS	LIVE	bridge.py:1318→autonomous_orchestrator	strategies.toml	NO
Orchestrator S20	Cross-Market Momentum	LIVE	bridge.py:1318→autonomous_orchestrator	strategies.toml	NO
IBS Mean Reversion	IBS_MR	LIVE	bridge.py:1342	Inline	NO
VolExpansion	VolExp	LIVE	bridge.py:1370	Inline	NO
ORB Breakout	ORB	LIVE	bridge.py:1401	Inline	NO
GapFade (inline)	GapFade	LIVE	bridge.py:1440	Inline	NO

[redacted] (DipRecovery)	[redacted]	[redacted] ONLY	[redacted]	[redacted]	[redacted]
[redacted] (EarlyRunner)	[redacted]	[redacted] ONLY	[redacted]	[redacted]	[redacted]
TypeC (OverboughtFade)	TypeC	**CLASSIFICATION ONLY**	bridge.py:633	config.toml:433	NO
TypeD (SupportBounce)	TypeD	**CLASSIFICATION ONLY**	bridge.py:643	config.toml:435	NO
TypeE (IBS)	TypeE	**CLASSIFICATION ONLY**	bridge.py:621	config.toml:441	YES
TypeF (OBVDivergence)	TypeF	**CLASSIFICATION ONLY**	bridge.py:617	config.toml:445	NO

Key insight: TypeA-F are CLASSIFICATION labels applied to signals in Stage 4 of bridge.py. They are NOT signal generators themselves. The 6 signal sources (VanguardSniper, Orchestrator, IBS, VolExp, ORB, GapFade) generate signals, then classify_entry_type() labels them. The Rust entry_engine.rs TypeA-F detectors are LIBRARY CODE not called at runtime.

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8. Strategy Sync Matrix

Strategy	In bridge.py?	In strategies.toml?	In config.toml?	In Rust?	In backtest?	In nightly?	In reporting?	Sy
[redacted]	[redacted] (source 1)	[redacted]	[redacted] config.py	[redacted]	[redacted] (backtest pipeline)	[redacted] (per-strategy)	[redacted]	OK
[redacted]	[redacted] (source 2)	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	**PA** stra not
IBS_MR	YES (source 3)	YES (ibs_mean_reversion)	NO	Rust has TypeE detector (unused)	NO	NO	NO	**PA**
VolExpansion	YES (source 4)	NO	NO	NO	NO	NO	NO	**MI** no tra
ORB	YES (source 5)	NO	NO	NO	NO	NO	NO	**MI** no tra
GapFade	YES (source 6)	YES (gap_fade)	NO	NO	NO	NO	NO	**PA**

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9. Live / Shadow / Disabled Boundary Register

Component	Boundary	Current state	Notes
Rust engine	LIVE (paper mode)	Running, processing ticks	IS_LIVE=false hardcoded

[redacted]	[redacted]	[redacted]	[redacted]
RiskArbiter	LIVE	27 CHECKs active	Many CHECKs skipped in sim mode
Chandelier exits	LIVE	Rung tracking active	No time-stop despite config
Claude curator	**SHADOW**	Evaluates signals, logs only, never blocks	Fail-open if Claude unavailable
Gemini scanner	**DISABLED** (silently)	API key commented out in .env	Cron runs but fails
Ouroboros learning	**FROZEN**	observe_only=true	Waits for N=300
config_writer	**FROZEN**	observe_only=true, exits early	dynamic_weights.toml stale from Mar 19
Nightly pipeline	LIVE	Runs 04:50 UTC daily	Analysis-only mode
Telegram alerts	LIVE	Kill switch, heartbeat, reports	Working
Google Sheets sync	LIVE	Every 5 min via cron	Working
IBKR scanner	**CONFIGURED**	config says enabled=true	Unclear if actually producing results
Backfill simulator	LIVE	07:00 UTC daily	Pre-market learning
Claude forensic review	LIVE	04:53 UTC daily	Post-nightly analysis

10. Model Role Matrix (Claude / Gemini)

Role	Model	Claude?	Gemini?	Reads from	Writes to	Classification	Wired?	Oper
Signal challenge	Claude	YES	NO	Signal dict + market [redacted]	claude_curator.ndjson	Shadow only	YES (bridge.py [redacted]	YES block
Forensic review	Claude	YES	NO	WAL trades + vetoes	claude reviews JSON	Cold path	YES (04:53 cron)	YES
Morning briefing	Claude	YES	NO	Nightly output + market data	Telegram message	Cold path	YES (07:45 cron)	YES
Evening briefing	Claude	YES	NO	Day's trades + P&L	Telegram message	Cold path	YES (21:30 cron)	YES
Gate calibration	Claude	YES	NO	Rejected trades WAL	Telegram + JSON	Cold path (weekly)	YES (Friday 22:00)	YES
Psych audit	Claude	YES	NO	Session performance	Telegram	Cold path (weekly)	YES (Sunday 23:00)	Ques value
Filing scanner	Claude	YES	NO	SEC/RNS feeds	Filing alerts	Shadow	YES (06:00 daily)	Unpr
[redacted] curation (core)	[redacted]	[redacted]	[redacted]	[redacted] + indicators	[redacted]	[redacted]	[redacted] (every 2h cron)	**BRO** API k .env

Universe curation (brief)	Gemini	NO	YES	Pre-market analysis	gemini_scanner.log	Cold path	YES (06:00 daily)	**BRO** same
Ticker selection	Neither	NO	NO	yfinance + contracts.toml	active_watchlist.json	Cold path	YES (every 2h)	YES (dete

Failure behaviors: - Claude unavailable: fail-open with 10% conservative haircut on confidence - Gemini unavailable: silently fails, ticker_selector still works (deterministic fallback)

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11. Current Deployment State

Component	Version/Commit	Location	Status
Rust binary	`d17d9bd`	/usr/local/bin/aegis	Running (PID 1 via tini)
Python bridge	`d17d9bd`	/app/python_brain/bridge.py	Running as subprocess
IB Gateway	gnzsnz/ib-gateway:stable	Container aegis-ib-gateway	Healthy, 4003 exposed
Redis	redis:7-alpine	Container aegis-redis	Healthy, AOF persistence
Config	`d17d9bd`	/app/config/	Baked into Docker image
WAL	Schema v1	/app/events/ (Docker volume)	7 archive files + current
system_memory	Last updated 2026-03-23 04:50 UTC	/app/data/system_memory.json	PF=0.0 (bug, fix deployed)

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12. Current Pipeline and Scheduling State (35 Cron Jobs)

Nightly Window (04:30-05:35 UTC)

Time	Job	Status
04:30	Docker prune	Best-effort (no Docker socket in container)
04:40	Parquet orphan cleanup	Active
04:45	Log rotation (7-day retention)	Active
04:50	**Nightly pipeline** (flock'd)	Active — nightly_v6 → config_writer → etc.
04:52	FTT registry (Monday only)	Active
04:53	Claude forensic review (flock'd behind nightly)	Active
04:54	Fill quality report (flock'd)	Active
04:55	Post-trade diagnostics (flock'd)	Active
05:00	Bridge stderr log rotation	Active
05:10	Ouroboros monitor (TOML health check)	Active
05:30	Universe updater (Wikipedia scrape)	Active
05:35	Universe sync (universe.json → contracts.toml)	Active

Market Hours

Time	Job	Status
Every 5 min	External monitor (deep health check)	Active
Every 5 min	Google Sheets sync	Active
Every 15 min (08-20)	Bridge health monitor	Active
Every 2h	ticker_selector (universe rotation)	Active
Every 2h (+5min)	Gemini core universe scan	BROKEN — no API key
Every 4h	Telegram heartbeat	Active
Every 6h	FX rate refresh (yfinance → fx_rates.toml)	Active
Every 6h	Contract expander	Active

Session Briefings (DST-aware dual-cron)

Time (London)	Session	Status
00:55	Asian	Active
06:00	Gemini morning brief	BROKEN
06:00	ISA universe refresh	Active
06:00	Claude filing scanner	Active
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
08:00	Daily health report	Active
14:25	American	Active
[redacted]	[redacted]	[redacted]
21:15	Daily sim trade report	Active
21:20	Cost drag report	Active
21:30	Claude evening briefing	Active

Weekly

Time	Job	Status
Sunday 22:00	IBKR full universe scanner	Active
Friday 22:00	Claude gate calibration	Active
Sunday 23:00	Claude psych audit	Active

13. Artifact Flow Map

```
INBOUND DATA:
  IBKR Gateway → Rust engine (ticks)
  yfinance → ticker_selector (prices)
```

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  Gemini API → gemini_scanner (curation) [BROKEN]
  Claude CLI → claude_curator (shadow verdicts)

INTERNAL FLOW:
  Rust engine → Python bridge (JSON stdin) → signal → Rust engine (JSON stdout)
  Rust engine → WAL (events/current.ndjson)
  WAL → nightly_v6 (analysis) → ouroboros_recommendations.json
  recommendations → config_writer → dynamic_weights.toml [FROZEN]
  ticker_selector → active_watchlist.json → Rust engine (file watch + resub)

OUTBOUND:
  Rust engine → Telegram (via WAL watcher + kill switch)
  Nightly → Telegram (reports, briefings)
  Nightly → Google Sheets (trade data)
  WAL → system_memory.json (cumulative stats)
```

[redacted]

14. Config Truth Map

Config file	Location	Generated by	Read by	Hot-reload?
config.toml	config/	Manual	Rust config_loader	At startup only
config.live.toml	config/	Manual	Rust config_loader (N8a overlay)	At startup only
contracts.toml	config/	contract_expander + manual	Rust + Python	SIGHUP
dynamic_weights.toml	config/	config_writer (FROZEN)	Rust + bridge.py	SIGHUP
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
active_watchlist.json	config/	ticker_selector	Rust (file mtime watch)	Yes (file watch)
initial_universe.toml	config/	ticker_selector	Rust config_loader	At startup only
fx_rates.toml	config/	fx_refresh cron	Rust	SIGHUP
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

[redacted]

STALE CONFIG: dynamic_weights.toml generated 2026-03-19 with WR=79.2% from 20 trades. Reality: WR=35.4% from 48 trades. The confidence_floor=65 set in this file is vetoing all current signals.

15. Parameter Truth Map

Parameter	config.toml value	dynamic_weights value	Runtime effective	Correct?
confidence_floor	55	65	65 (dynamic overrides)	**WRONG** — too high for current strategies
max_positions	999 (simulation section)	N/A	999	OK for paper
chandelier_atr_mult	3.0 (default)	3.05	3.05	OK
spread_veto_pct	0.3%	N/A	0.3%	OK
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
daily_trade_limit	999999 (simulation)	N/A	999999	OK for paper

[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
observe_only [redacted]	true [redacted]	N/A [redacted]	true [redacted]	Intentional (needs [redacted]
exit_time_stop.enabled	true	N/A	**NOT IMPLEMENTED**	**BUG**
exit_time_stop.max_minutes_to_rung2	45	N/A	**NOT IMPLEMENTED**	**BUG**

16. Contradiction Register

ID	Subsystem	Files	Docs say	Code says	Runtime says	Severity
C01 [redacted]	Risk CHECKs [redacted]	risk_arbiter.rs, [redacted]	"33 CHECKs" [redacted]	**27 active CHECKs** (3,4 [redacted] removed)	27 [redacted]	LOW
C02	TypeB signals	bridge.py, backtest	"52.4% WR, best strategy"	classify_entry_type needs 3-bar rising RVOL	0 TypeB trades ever	HIGH
C03	entry_engine.rs	entry_engine.rs, tasks/todo.md	"Dead code" / "Wire TypeA-F"	786 LOC compiles, NOT called at runtime	Never executes	MEDIUM
[redacted]	[redacted]	[redacted] exit_engine.rs	[redacted] `= true`	**FIXED**: time-stop now in [redacted] (45min→0.3x ATR trail)	Deployed 2026-03-24	**FIXED**
C05 [redacted]	Confidence floor [redacted]	config.toml, [redacted]	"config=55" (wrong — was [redacted]	BOTH config.toml [redacted] had 65	ALL [redacted] vetoed at conf=49	**FIXED**
C06	Gemini	.env, entrypoint.sh	"API key SET" (manual says)	**FIXED**: uncommented in .env, pre-flight warns if missing	Deployed 2026-03-24	**FIXED**
C07 [redacted]	Active tickers [redacted]	config, logs [redacted]	"50 tickers loaded" [redacted]	50 attempted, tick- [redacted]	Only 11 [redacted] data	HIGH [redacted]
C08	PF calculation	persistent_memory.py	"PF should be computed"	Never computed (was 0.0)	Fixed this session	CLOSED
C09	Paper fills	master plan v1	"Possible mid-point fills"	Fills at ask/bid (realistic)	Confirmed ask entry, bid exit	CLOSED
C10	dynamic_weights	dynamic_weights.toml	"Generated nightly"	observe_only=true → exits early	**STALE since Mar 19**	HIGH

[redacted]

17. Dead Code / Dead Path Register

Item	Location	LOC	Status	Why dead	Action
entry_engine.rs [redacted]	rust_core/src/ [redacted]	787	Compiles, [redacted]	Python bridge handles all [redacted]	Add [redacted] comment
rotation_scanner.rs	rust_core/src/ rotation_scanner.rs	~200	Instantiated, never called	HotScanner does apex ticks; RotationScanner idle	Leave (may wire later)
PaperBroker for sim fills	rust_core/src/ paper_broker.rs	453	Used for tests only	simulation_mode bypasses broker entirely	Keep for tests
Docker prune cron	crontab line 30	1	Runs but no-op	No Docker socket inside container	Remove or fix
Gemini scanner crons	crontab	3	Run but fail [redacted]	GEMINI_API_KEY [redacted]	Fix .env or [redacted]

[redacted]

Document	Location	Status	Why stale
MERGED_MASTER_PLAN_v1.0	docs/ archive/	SUPERSEDED	Replaced by this plan
PLAN_1_ENGINE_FIX_AND_MULTI_EXCHANGE.md	project root	SUPERSEDED	Sprints 0-10 completed, new sprints here
[redacted]	[redacted]	[redacted]	[redacted] items remain
[redacted]	[redacted] root	[redacted]	[redacted]
108 docs in docs/archive/	docs/ archive/	ARCHIVED	Historical context only
[redacted]	[redacted]	[redacted]	[redacted] old contradictions, [redacted] resolved
dynamic_weights.toml	config/	STALE	Generated Mar 19, WR=79.2% vs reality 35.4%

19. Risk / Safety Control Map

Control	Implementation	Active?	Bypassed in sim?
IS_LIVE=false hardcoded	main.rs	YES	N/A
27-CHECK risk arbiter	risk_arbiter.rs	YES	8 CHECKs skipped in sim (6,14,15,16,17,18,30,31,32)
Chandelier 5-rung stop	exit_engine.rs	YES	No
EOD flatten (16:25 London)	exit_engine.rs	YES	No
[redacted] files)	[redacted]	[redacted]	[redacted]

Consecutive loss breaker	risk_arbiter.rs CHECK 21	YES	No
Daily drawdown halt	risk_arbiter.rs CHECK 18	YES	Skipped in sim
Zombie halt recovery	risk_arbiter.rs	YES (30-min timeout)	No
Bridge watchdog	entrypoint.sh + bridge_watchdog.py	YES	No
IBKR circuit breaker	broker_circuit_breaker in engine	YES	No
Ouroboros freeze	config.toml observe_only	YES	N/A
Claude shadow-only	claude_curator.py	YES	N/A

20. Pre-Live Blockers (MUST fix before real capital)

#	Blocker	Why	Sprint	Effort
1	IS_LIVE=false hardcoded	Cannot trade live	Rust change	5 min
2	8 paper overrides in simulation section	max_positions=999, heat=100%, etc.	Config revert via config.live.toml	15 min
3	**No time-stop** despite config saying enabled	Capital locked in sideways positions	Sprint S3	2h
4	WR 35.4% (need 40%)	Below validation gate	More trades + strategy improvement	Weeks
5	PF ~0.77 (need 1.3)	Below validation gate	Strategy improvement	Weeks
6	14 consecutive losses (need <8)	Below validation gate	Better entry filtering	Weeks
7	Only 1 of 6 strategies proven	Insufficient diversification	Run 300+ trades	Weeks
8	Ouroboros frozen on N=48	ML loop needs N=300	Wait for trades	Weeks
9	EC2 4GB RAM	OOM risk under stress	Sprint S5: upgrade	15 min
10	Disk at 80%	Cannot rebuild Rust	Prune + expand	15 min
11	Zero slippage simulation	P&L optimistic	Add slippage_bps config	1h

21. Post-Live Non-Blockers (Safe to do after live)

Item	Why post-live	Priority
Claude curator → veto authority	Needs validation data first	Medium
Gemini → signal-level integration	Currently universe-only	Low
Advanced regime detection (VPIN hot-path)	Marginal improvement	Low
Rust entry_engine.rs wire-up	Python classification works fine	Low
Multi-account ISA/GIA split	Single account first	Low
Advanced order types (iceberg, TWAP)	Simple limit orders fine initially	Low

22. Daily Operating Workflow

Time (UTC)	Action	Command/Check
07:00	Check containers healthy	`ssh EC2 'docker ps'`
07:00	Check overnight errors	`docker logs aegis-v2 --tail 30`
07:00	2FA if Monday	IBKR mobile app
07:45	Read Claude morning briefing	Telegram
08:00	LSE open: verify tick flow	`grep SIGNAL_ARRIVED` in logs
[redacted]	[redacted]	[redacted]
14:30	US open: check for ORB/momentum	Logs
[redacted]	[redacted]	[redacted]
21:00	Read daily sim trade report	Telegram
21:15	Read cost drag report	Telegram
21:30	Read Claude evening briefing	Telegram

[redacted]

Event	Action	Automated?
Signal arrives	Bridge generates, RiskArbiter evaluates	YES
Signal vetoed	Logged to WAL + telemetry	YES
[redacted]	[redacted]	[redacted]
Rung advance	Chandelier stop ratcheted up	YES
EOD flatten	All positions closed at 16:25 London	YES
Bridge crash	Watchdog detects (30s), restarts	YES
IBKR disconnect	Engine reconnects (10 attempts, backoff)	YES
[redacted]	[redacted]	[redacted]

24. End-of-Day / LSE Close Workflow

Time (London)	Event	Owner
16:25	EOD flatten trigger fires	Rust exit_engine
16:25	All open positions get MarketSell exit	Rust engine
16:30	LSE closing auction — no new entries	Rust blackout window
16:30	US-only session briefing sent	Telegram cron
17:00	Dark mode begins (no new entries)	Rust engine

20:00	ModeC (US-session) may allow entries	Rust engine
21:00	US close — final exits	Rust engine
21:15	Daily sim trade report	Python cron

[redacted]

25. Nightly Workflow

See Section 12 for complete schedule. Key sequence: 1. **04:50** — nightly_v6.py: load WAL trades → compute metrics → recommendations → update persistent_memory 2. **04:51** — config_writer: reads recommendations → generates dynamic_weights.toml (FROZEN — observe_only) 3. **04:53** — Claude forensic review: classifies trades + vetoes 4. **04:54** — Fill quality report: paper→live slippage estimate 5. **04:55** — Post-trade diagnostics: consolidated analysis 6. **05:10** — Ouroboros monitor: TOML health + staleness check

[redacted]

26. Recovery / Restart / Rollback Workflow

[redacted]

Kill Switch

```
ssh EC2 'docker exec aegis-v2 touch /app/KILL'      # Halt trading
ssh EC2 'docker exec aegis-v2 touch /app/PAUSE'     # Flatten + halt
ssh EC2 'cd ~/nzt48-aegis-v2 && docker compose down' # Full stop
```

[redacted]

Restart

[redacted]

```
ssh EC2 'docker exec aegis-v2 rm -f /app/KILL /app/PAUSE'
ssh EC2 'cd ~/nzt48-aegis-v2 && docker compose up -d'
```

Rollback to Previous Commit

[redacted]

```
git revert HEAD && git push
rsync ... && ssh EC2 'docker compose build aegis-v2 && docker compose up -d && docker system prune -f'
```

Full Rebuild (Python changes)

```
git push && rsync ... && ssh EC2 'docker system prune -f && docker compose build aegis-v2 && docker com
```

[redacted]

Full Rebuild (Rust changes — needs >5GB free)

[redacted]

```
ssh EC2 'docker system prune -af'  # Aggressive prune
rsync ... && ssh EC2 'docker compose build --no-cache aegis-v2 && docker compose up -d'
```

27. Validation and Readiness Gates

[redacted]

100-Trade Rolling-Window Gates

Gate	Threshold	Current (N=48)	Status
[redacted]	[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]	[redacted]
Max Consecutive Losses	< 8	14	FAILING
Chandelier Rung >= 2	>= 50% of exits	Unknown	UNMEASURED
Max Daily Drawdown	< 3%	OK	PASSING
Strategy Diversity	>= 2 strategies with WR > 35%	1 only	FAILING

Pre-Validation Immediate Fixes Needed

1. **CRITICAL**: Lower confidence_floor in dynamic_weights to allow signals through
2. **CRITICAL**: Fix IBKR tick-by-tick subscription limit (reduce from 102 L1 + 200 mkt)
3. **HIGH**: Uncomment GEMINI_API_KEY in .env for universe curation

[redacted]

28. ROI-Ranked Action Backlog

Priority	Action	ROI	Sprint
[redacted]	[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]	S-HOTFIX
[redacted]	[redacted]	[redacted]	[redacted]
HIGH	Uncomment GEMINI_API_KEY in .env	Universe curation broken	S-HOTFIX
HIGH	Investigate TypeB threshold mismatch	Best backtest strategy never fires	S4
[redacted]	[redacted]	[redacted]	S5
MEDIUM	Create config.live.toml with safe production values	Deployment safety	S6
MEDIUM	Add slippage simulation to paper fills	P&L realism	S7
LOW	Quarantine entry_engine.rs dead code	Code clarity	S3 (bundle)
LOW	Verify Gemini scanner produces valid output	Gemini integration	S8

29. Chunked Implementation Sprints

Sprint S-HOTFIX: Unblock Signal Flow (30 min, CRITICAL)

- **Priority**: CRITICAL
- **Classification**: Fix Now
- **Why**: System is generating signals but ALL are vetoed. Zero trades being made. The paper trading validation gate cannot advance.
- **Files**: `config/dynamic_weights.toml`, `.env`
- [redacted]
- Set `confidence_floor = 45` in dynamic_weights.toml (was 65, killing everything)
- Uncomment `GEMINI_API_KEY` in `.env`
- Reduce L1 subscription attempts (or accept graceful degradation)

- [redacted]
- **Success criteria**: Signals no longer all vetoed. At least 1 trade per day.
- **Rollback**: Revert dynamic_weights.toml to previous values

[redacted]

- **Priority**: Critical
- **Classification**: Build Now
- **Why**: Config says `exit_time_stop.enabled=true, max_minutes_to_rung2=45` but exit_engine.rs has no time-stop code. Capital locks in sideways positions until EOD.
- **Files**: `rust_core/src/exit_engine.rs`, `rust_core/src/config_loader.rs`
- **Changes**: Add time-stop logic: if position held > max_minutes and not at rung 2+, tighten trail to aggressive_trail_atr
- **Blocker**: Needs >5GB free disk for Rust rebuild. EC2 at 80% — prune first.
- [redacted]
- **Rollback**: Set exit_time_stop.enabled=false in config.toml
- **Pre-live**: MANDATORY

[redacted]

- **Priority**: High
- **Classification**: Build Now
- **Why**: TypeB has 52.4% WR in backtest but 0 production signals. Best theoretical strategy is silent.
- **Files**: `python_brain/bridge.py:624-630` (classify_entry_type)
- [redacted]
- **Test**: Add gate veto logging for TypeB threshold checks
- [redacted]

Sprint S5: EC2 Instance Upgrade (15 min)

- **Priority**: High
- [redacted]
- **Method**: AWS console → Stop instance → Change type → Start → 2FA re-auth
- **Risk**: 5 min downtime
- **Success criteria**: `free -m` shows 8GB
- **Rollback**: Change back to c7i-flex.large

Sprint S6: config.live.toml (15 min)

- [redacted]
- **Why**: 8 paper overrides would be catastrophic in live mode
- **Files**: `config/config.live.toml` (already exists with N8a overlay)
- [redacted]

---

30. Deployment Sequence

For any change: 1. `git add` specific files → `git commit -m "..."` → `git push` 2. `rsync -avz --exclude='.git' --` [redacted] `build aegis-v2 && docker compose up -d` 4. If Rust change: `docker system prune -f` first, then build 5. Verify: `docker ps`, `docker logs aegis-v2 --tail 30`, check for errors

---

[redacted]

[redacted]
for SIGNAL_ARRIVED in logs — bridge generating signals 4. Check for VETO vs SIM_TRADE — signals getting through
[redacted]

32. Success Criteria

Milestone	Metric	Target	Current
Signal flow unblocked	Signals not all vetoed	>0 trades/day	0 (all vetoed)
[redacted]			
Validation gate WR	Win rate	>= 40%	35.4%
Validation gate PF	Profit factor	>= 1.3	~0.77
Validation gate losses	Max consecutive	< 8	14
Strategy diversity	Strategies with trades	>= 2	1
Live readiness	All pre-live blockers resolved	0 blockers	11 blockers

33. Failure / Escalation Rules

Condition	Action	Authority
Daily drawdown > 3%	Automatic FLATTEN	Rust risk_arbiter
3 consecutive stop losses	Automatic HALT	Rust risk_arbiter
[redacted]		
IBKR disconnect > 60s	Auto-reconnect with backoff	Rust engine
IBKR disconnect > 10min	HALT mode	Rust engine
Disk > 90%	ALERT operator (Telegram)	external_monitor
Kill switch triggered	Flatten all, halt	Telegram → kill_switch.py
Equity < 70% of initial	Automatic HALT (CHECK 32)	Rust risk_arbiter

34. Final Canonical Plan Verdict

[redacted] grade risk infrastructure in Rust. As of this update (v2.1), the critical runtime blockers have been FIXED AND DEPLOYED:

Issue	Status	Fix
All signals vetoed (confidence_floor=65)	FIXED	config.toml lowered to 50, dynamic_weights to 45
[redacted]		
Gemini API key missing	FIXED	Uncommented in .env, deployed
PF=0.0 bug	FIXED	persistent_memory.py now tracks cumulative gross wins/losses

A2. Data Asphyxiation (Critique: system operated blind to 78% of universe)

Response: 100% valid. IBKR tick-by-tick limit (100 lines) means requesting 302 subscriptions is guaranteed to fail silently. The system should detect subscription integrity and adjust.

Action: Acknowledged as known limitation. The engine currently logs SUB ERROR for failed L1 subs but continues operating on the successful ones. MktData (snapshot) subscriptions succeed for all 100 requested. For paper validation, operating on ~100 MktData feeds is sufficient. Subscription integrity check added to pre-live blockers list.

A3. Time-Stop During Halts (Critique: naive clock-based time-stop will fire during exchange halts)

Response: Valid concern, mitigated by design. The time-stop fires ONLY when `evaluate()` is called AND `current_price <= aggressive_stop`. During an exchange halt, no ticks arrive, so evaluate() is never called. After unhalt, the aggressive stop is already set and the market decides if it fires — not a forced instant exit.

Action: Time-stop implemented with halt-safety by design. TODO added for active-trading-minute counter for live mode.

A4. TypeB Backtest Illusion (Critique: 3-bar rising RVOL is a data artifact)

Response: Correct diagnosis. TypeB classification requires 3 consecutive ticks of strictly increasing RVOL, which is rare in real-time where RVOL is a slow-moving 20-bar ratio. The backtest likely benefited from bar-aggregated data where RVOL changed more discretely.

Action: Diagnostic logging added. If TypeB never fires after 1 week of lowered confidence floor, the classification threshold will be relaxed.

A5. Pre-Flight Safety (Critique: system should crash-on-boot for missing deps)

Response: 100% valid.

Action: Pre-flight dependency checks added to entrypoint.sh. Critical failures (missing config, missing Redis password, disk >90%) now crash the container. Optional deps (Gemini, Telegram) produce loud warnings.

Appendix B: Execution Roadmap (Institutional Syndicate + ChatGPT Integrated)

Build Now (Tier 1) — DONE or IN PROGRESS

#	Action	Status	Evidence
1	Canonical strategy registry	DONE	`config/strategy_registry.json` created
2	TypeB + VanguardSniper as only LIVE strategies	DONE	bridge.py enforces: TypeA/D disabled, TypeC/E/F shadow
3	Hard-shadow non-core strategies	DONE	bridge.py returns None for disabled/shadow types
4	Quarantine dead Rust entry_engine.rs	DONE	QUARANTINE NOTICE added, imports preserved for compilation
5	Inject 4 academic frameworks into Claude prompts	DONE	López de Prado, Hasbrouck, Almgren/Chriss, Bollerslev injected
6	Confidence floor fix (root cause in Python bridge)	DONE	Hurst threshold H<0.50→H<0.30, volume floor 75→60
7	Time-stop with halt-safe active-trading-ticks	DONE	exit_engine.rs uses tick counter, not wall clock
8	Validation counter reset (clean post-patch)	DONE	system_memory zeroed, pre-patch archived

9	Pre-flight dependency checks	DONE	entrypoint.sh crashes on missing deps
10	Cron optimization (35→32 jobs)	DONE	Dead/duplicate crons disabled, frequency reduced
11	Regime routing skeleton	DONE	strategy_registry.json has regime_allowed/blocked per [redacted]
12	Session templates skeleton	DONE	strategy_registry.json has session_allowed/blocked per strategy

Build Now (Tier 1) — REMAINING

#	Action	Priority	Effort
1	Wire strategy_registry.json into nightly reporting	HIGH	2h
2	Sizing hardening (layered: regime × symbol × session × drawdown)	HIGH	4h
3	Symbol-quality memory (per-symbol spread/slippage/WR tracking)	HIGH	4h
[redacted]	[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]	[redacted]
6	Nightly trade packet upgrade (full WAL enrichment)	MEDIUM	2h

[redacted]

#	Action	Notes
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
3	New intraday candidates	ORB, GapFade, VolExpansion — log signals only
4	Claude/Gemini cold-path enhancements	Filing triage, anomaly grouping
[redacted]	[redacted]	[redacted]

Delete Now

#	Action	Files
1	Dead Docker prune cron	crontab (DONE — disabled)
[redacted]	[redacted]	[redacted]
3	TypeA/D live trading path	bridge.py (DONE — returns None)
4	Stale dynamic_weights WR=79.2%	Will self-correct after nightly on new trades

[redacted]

#	Action	Condition
1	Model hot-path authority	Never — deterministic risk only
[redacted]	[redacted]	[redacted]
3	More venues	Must prove spread + fill quality first
4	Full Kelly sizing	Requires PF > 1.5 sustained over 300+ trades